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Litterature:

H. H. Hamedani, *Probability and Statistics: The Science of Uncertainty*, 3rd ed.,  
<https://www.probabilitycourse.com>

## Applications of Probability and statistics

### Signal Processing (DSP)

- Modeling of noise and random signals
- Stochastic processes in time and frequency
- Spectral estimation
- Wiener and Kalman filtering
- Detection and estimation theory
- Sensor fusion
- Time–frequency analysis of random signals

### Machine Learning and Artificial Intelligence

- Learning from random samples
- Generalization and uncertainty analysis
- Probabilistic models for data
- Bayesian inference
- Gaussian processes
- Stochastic optimization

### Communications Engineering

- Noise modeling in transmission channels
- Error probability analysis
- Information theory
- Channel capacity
- Fading and interference models
- Coding and decoding under uncertainty

## Physics

- Statistical mechanics
- Thermodynamics (microscopic interpretation)
- Quantum mechanics (measurement and expectation values)
- Brownian motion and diffusion

## Finance and Economics

- Risk modeling
- Asset price dynamics
- Stochastic differential equations
- Portfolio optimization
- Econometric modeling
- Time series analysis

## Biology and Medicine

- Population dynamics
- Epidemiological models
- Survival analysis
- Clinical trial design
- Neural spike train modeling

## Computer Science

- Randomized algorithms
- Probabilistic data structures
- Monte Carlo methods
- Average-case complexity analysis
- Error analysis in numerical algorithms

## **Reliability and Engineering Systems**

- Failure-time modeling
- Lifetime distributions
- Risk and safety analysis
- Maintenance optimization
- Fault detection

## **Image Processing and Computer Vision**

- Noise modeling
- Image restoration
- Bayesian inverse problems
- Markov random fields
- Uncertainty in reconstruction

## **Climate and Environmental Modeling**

- Stochastic climate models
- Random fields
- Spatio-temporal statistics
- Extreme value theory
- Uncertainty in long-term prediction

## **Engineering Measurement and Metrology**

- Measurement uncertainty
- Error propagation
- Sensor calibration
- Data fusion from multiple instruments

## Review of Set Theory

- A **set** is a collection of elements (e.g., numbers, symbols).

Example:  $A = \{\clubsuit, \diamond\}$

- **Notation:**

- $\in$ : "belongs to" (e.g.,  $\diamond \in A$ )
- $\notin$ : "does not belong to" (e.g.,  $\heartsuit \notin A$ )
- **Order does not matter** in sets:  $\{a, b\} = \{b, a\}$

### Common Sets Used:

- $\mathbb{N}$ : Natural numbers  $\{1, 2, 3, \dots\}$
- $\mathbb{Z}$ : Integers  $\{\dots, -2, -1, 0, 1, 2, \dots\}$
- $\mathbb{Q}$ : Rational numbers (fractions)
- $\mathbb{R}$ : Real numbers
- $\mathbb{C}$ : Complex numbers  $a + bi$ , where  $i = \sqrt{-1}$

### Set of numbers in an interval:

- **Closed:**  $[2, 3] \rightarrow 2 \leq x \leq 3$
- **Open:**  $(-1, 3) \rightarrow -1 < x < 3$
- **Half-open:**  $[1, 2) \rightarrow 1 \leq x < 2$  ..

## Review of Set Theory

Sets can also be defined by properties:

$$A = \{x \mid x \text{ satisfies some property}\}$$

The symbol “ $\mid$ ” means “such that”.

### Examples:

- $C = \{x \in \mathbb{Z} \mid -2 \leq x < 10\} = \{-2, -1, \dots, 9\}$
- $D = \{x^2 \mid x \in \mathbb{N}\} = \{1, 4, 9, 16, \dots\}$

## Subsets

- $A \subset B$  means every element of  $A$  is in  $B$
- $B \supset A$  is the converse ( $B$  contains  $A$ )

### Examples:

- $\{1, 4\} \subset \{1, 4, 9\}$  or  $\{1, 4\} \subseteq \{1, 4\}$
- $\mathbb{N} \subset \mathbb{Z}, \mathbb{Q} \subset \mathbb{R}$

## Equality of Sets

- Two sets are equal if they have the **same elements**

Example:  $\{1, 2, 3\} = \{3, 2, 1\}$

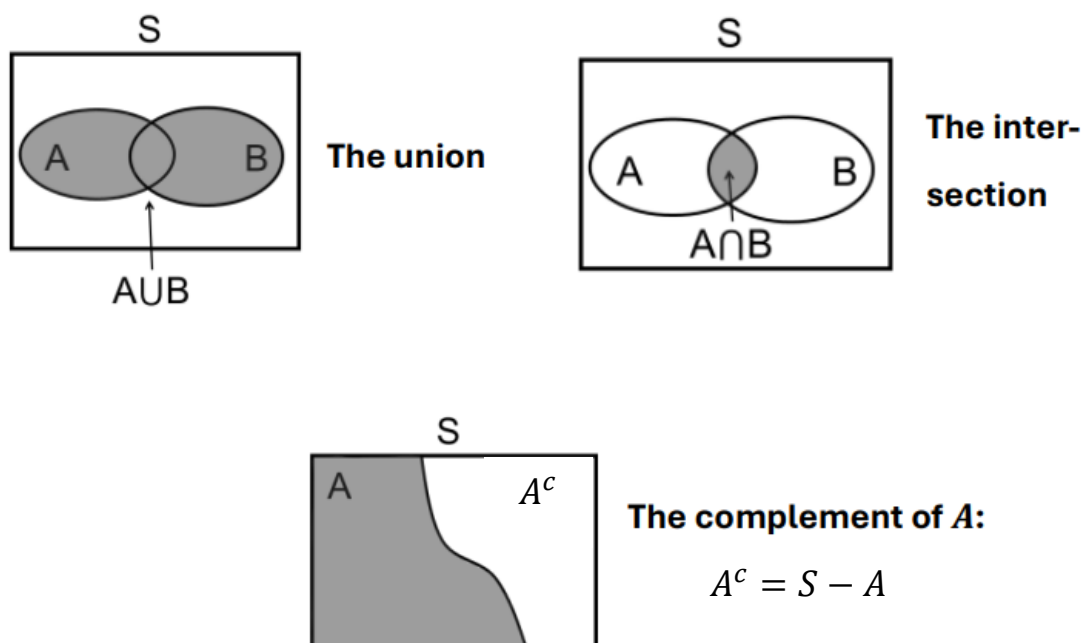
## Review of Set Theory

### Special Sets

- **Empty set:**  $\emptyset$ , has no elements.  $\emptyset \subset A$  for any set  $A$
- **Universal set:** Set of all possible outcomes in a given context (denoted  $S$ )
  - Example: Tossing a coin  $\rightarrow S = \{H, T\}$
  - Rolling a die  $\rightarrow S = \{1, 2, 3, 4, 5, 6\}$

### Venn Diagrams

- Visualize how sets relate to each other.
- The **universal set**  $S$  is drawn as a rectangle; sets are circles.



## Review of Set Theory

### Power Set (Optional)

The **power set** of a set is the set of **all possible subsets**, including:

- the empty set  $\emptyset$
- the set itself
- every combination of its elements

### Example:

Let's say we have the set:

$$A = \{1, 2\}$$

Then the **power set** of  $A$ , written  $\mathcal{P}(A)$ , is:

$$\mathcal{P}(A) = \{\emptyset, \{1\}, \{2\}, \{1, 2\}\}$$

- It has  $2^n$  elements, where  $n$  is the number of elements in the original set.

### Universal set vs Power set

#### Tossing a Coin:

Let's say we flip a coin once.

- **Universal Set:**

$$S = \{H, T\}$$

→ This represents *all possible outcomes*.

- **Power Set of  $S$ :**

$$\mathcal{P}(S) = \{\emptyset, \{H\}, \{T\}, \{H, T\}\}$$

→ This lists *all subsets* of  $S$ .

## Review of Set Theory

### De Morgan's Laws

For any sets  $A_1, A_2, \dots, A_n$ :

- The complement of a union is the intersection of the complements:

$$(A_1 \cup A_2 \cup \dots \cup A_n)^c = A_1^c \cap A_2^c \cap \dots \cap A_n^c$$

- The complement of an intersection is the union of the complements:

$$(A_1 \cap A_2 \cap \dots \cap A_n)^c = A_1^c \cup A_2^c \cup \dots \cup A_n^c$$

### Example

Let:

$$A = \{1, 2\}, \quad B = \{2, 3\}, \quad C = \{3, 4\}, \quad U = \{1, 2, 3, 4\}$$

- $B \cup C = \{2, 3, 4\}$
- $(B \cup C)^c = U - \{2, 3, 4\} = \{1\}$

Now:

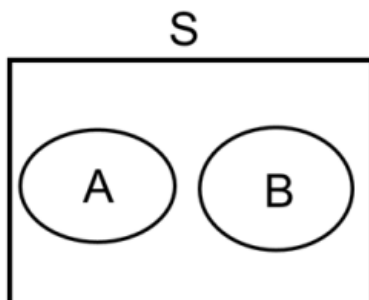
- $B^c = U - B = \{1, 4\}$
- $C^c = U - C = \{1, 2\}$
- $B^c \cap C^c = \{1, 4\} \cap \{1, 2\} = \{1\}$

So:

$$(B \cup C)^c = B^c \cap C^c = \{1\}$$

**Quiz**

Consider the venn diagram and determine which statement is true



**A:**  $A \cap B = \emptyset$

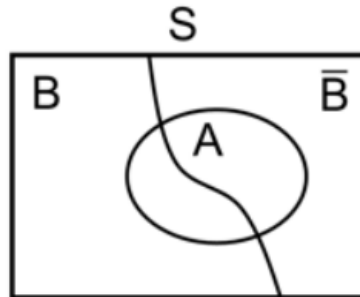
**B:**  $A \cap B = S$

**C:**  $A - B = \emptyset$

**D:**  $A \cup B = S$

**Quiz**

Consider the venn diagram and determine which statement is true



**A:**  $A \cup B \subset \bar{B}$

**B:**  $A \cap \bar{B} \subset B$

**C:**  $A \cup \bar{B} \subset B$

**D:**  $\bar{A} \cap S \subseteq \bar{A}$

## Cardinality

- Cardinality means the **number of elements** in a set.
- For a **finite set**, it's just the count of elements:

$$|A| = \text{number of elements in } A$$

### Example:

By definition, a set contains only unique elements:

$$A = \{2,4,4,6\} = \{2,4,6\}$$

So

$$|A| = 3$$

## Inclusion-Exclusion Principle (Finite Sets)

- For two sets:

$$|A \cup B| = |A| + |B| - |A \cap B|$$

### Example:

$$A = \{2,4,6\} \text{ and } B = \{2,8,10\}$$

So

$$|A \cup B| = 3 + 3 - 1$$

**Quiz**

Let  $A$  and  $B$  be two events such that

$$P(A) = 0.4, \quad P(B) = 0.7, \quad P(A \cup B) = 0.9.$$

What is  $P(A \cap B)$ ?

A. 0.1

B. 0.2

C. 0.3

D. 0.4

**Quiz**

At a coding bootcamp:

- 30 students are enrolled in the **Python** course (Set  $A$ )
- 25 students are enrolled in the **Data Science** course (Set  $B$ )
- 10 students are enrolled in **both** courses

How many **distinct students** are enrolled in **at least one** of the two courses?

- A) 65
- B) 45
- C) 55
- D) 70

## Cardinality

### Countable vs. Uncountable Sets

#### Countable:

- A set is countable if:
  1. It's finite, or
  2. It can be listed like  $\{a_1, a_2, a_3, \dots\}$  in 1-1 correspondence with  $\mathbb{N}$
- Examples:  
 $\mathbb{N}, \mathbb{Z}, \mathbb{Q}$  and all their subsets

#### Uncountable:

- Cannot be listed; no 1-1 mapping to  $\mathbb{N}$
- Example: Any interval in  $\mathbb{R}$ , e.g.  $[0, 1]$

**Quiz**

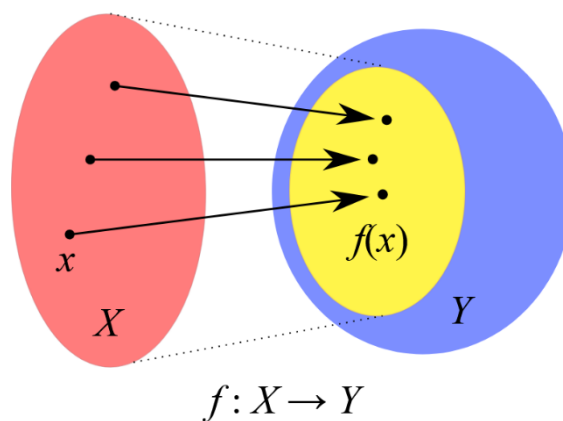
**Which of the following sets is countable?**

- A) The set of all complex numbers  $\mathbb{C}$
- B) The union of  $\mathbb{N}$  and  $\mathbb{Q}$
- C) The set of all real numbers that are not rational
- D) The set of all **irrational** numbers

## Functions

- A function  $f : A \rightarrow B$  maps each element in a **domain**  $A$  to exactly one element in a **co-domain**  $B$ .
- The **range** of a function is the set of all actual outputs  $f(x)$  and is always a **subset of the co-domain**.
- **Notation:**

$f : A \rightarrow B$  means input from  $A$ , output in  $B$



### Example

Let:

$$f : \mathbb{R} \rightarrow \mathbb{R}, \quad f(x) = x^2$$

- **Domain:**  $\mathbb{R}$  (all real numbers)
- **Co-domain:**  $\mathbb{R}$
- **Range:**  $\mathbb{R}^+ = \{x \in \mathbb{R} \mid x \geq 0\}$

## Basics of Probability

Probability theory **borrowes directly** from set theory — it's the **same machinery**, just applied to modeling randomness:

| Set Theory Term         | Probability Term             |
|-------------------------|------------------------------|
| Universal set $S$       | Sample space $\Omega$ or $S$ |
| Subset $A \subset S$    | Event $A \subset \Omega$     |
| Element $x \in A$       | Outcome occurs               |
| Complement $A^c$        | "Not A"                      |
| Union $A \cup B$        | "A or B" happens             |
| Intersection $A \cap B$ | "A and B" happens            |

### Example of Random Experiment

Roll a dice. Sample space  $S$ , of a single roll, is given by

$$S = \{1, 2, 3, 4, 5, 6\}$$

An **event** could be the even numbers:

$$A = \{2, 4, 6\} = \{2\} \cup \{4\} \cup \{6\}$$

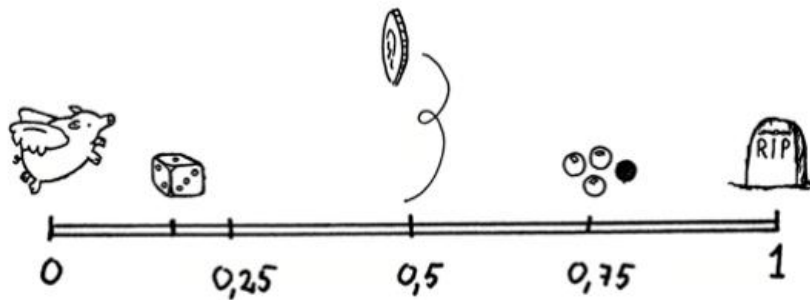
**Quiz**

**What is the sample space of rolling a single dice twice?**

- A.  $S = \{1, 2, 3, 4, 5, 6\}$
- B.  $S = \{(i, j) \mid i \in \{1, 2, 3, 4, 5, 6\}, j \in \{1, 2, 3, 4, 5, 6\}\}$
- C.  $S = \{(1, 1), (1, 2), \dots, (6, 6)\}$
- D.  $S = \{2, 3, \dots, 12\}$
- E.  $S = \{(i, j) \mid i + j = 7\}$

## Basics of Probability

- A probability  $P(A)$  is a number between 0 and 1 indicating how likely event  $A$  is.
  - $P(A) = 0$ : impossible
  - $P(A) = 1$ : certain
  - The closer to 1, the more likely the event



## Axioms of Probability

1. Non-negativity:

$$P(A) \geq 0$$

2. Normalization:

$$P(S) = 1$$

where  $S$  is the **sample space** (all possible outcomes)

3. Additivity (for disjoint events):

$$P(A_1 \cup A_2 \cup \dots) = P(A_1) + P(A_2) + \dots$$

if all events are **mutually exclusive**

## Basics of Probability

### Notation

- $P(A \cap B) = P(\text{A and B}) = P(A, B)$   
→ Probability that **both** events A and B occur.
- $P(A \cup B) = P(\text{A or B})$   
→ Probability that **at least one** of the events A or B occurs.

### Axioms of Probability method. Example:

Probability of **event**  $E = \{1, 5\}$  when rolling a fair die

1. **Given:** The die is **fair**, so all 6 outcomes are **equally likely**.

That means:

$$P(\{1\}) = P(\{2\}) = \dots = P(\{6\}) = p$$

2. **Axiom 2 (Normalization):**

The probability of the entire sample space is 1:

$$P(S) = P(\{1, 2, \dots, 6\}) = 1$$

3. **Axiom 3 (Additivity):**

Since the outcomes are disjoint:

$$P(S) = P(\{1\}) + P(\{2\}) + \dots + P(\{6\}) = 6p \Rightarrow p = \frac{1}{6}$$

4. So,

$$P(\{1\}) = P(\{5\}) = \frac{1}{6}$$

5. Now use **Axiom 3 again** for the event  $E = \{1, 5\}$ :

$$P(E) = P(\{1, 5\}) = P(\{1\}) + P(\{5\}) = \frac{1}{6} + \frac{1}{6} = \frac{1}{3}$$

## Basics of Probability

### Relative Frequency

If the outcomes are **equally likely**, the probability of any event  $E \subseteq S$  is:

$$P(E) = \frac{|E|}{|S|}$$

where:

- $|E|$ : the number of outcomes in the event
- $|S|$ : the total number of outcomes in the sample space

### Relative frequency method. Example:

Probability of  $E=\{1,5\}$  when rolling a fair die

- Sample space:  $S = \{1, 2, 3, 4, 5, 6\} \Rightarrow |S| = 6$
- Event:  $E = \{1, 5\} \Rightarrow |E| = 2$

$$P(E) = \frac{|E|}{|S|} = \frac{2}{6} = \frac{1}{3}$$

**Quiz**

Let  $S = \{1, 2, 3, 4, 5, 6\}$  be the sample space for rolling a fair six-sided die.

Let the events be:

- $A = \{1, 2, 3\}$  ("Roll is less than 4")
- $B = \{3, 4, 5\}$  ("Roll is between 3 and 5")

What is the probability  $\Pr(A \cap B)$

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A)  $\frac{3}{6}$

B)  $\frac{1}{6}$

C)  $\frac{2}{6}$

D)  $\frac{1}{3}$

**Quiz**

Let  $A$ ,  $B$ , and  $C$  be three events in a sample space  $S$ .

Suppose the following is known:

- $A \cup B \cup C = S$
- $P(A) = \frac{1}{2}$
- $P(B) = \frac{2}{3}$
- $P(A \cup B) = \frac{5}{6}$

What is  $P(A \cap B)$  ?

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- A)  $\frac{1}{3}$
- B)  $\frac{1}{6}$
- C)  $\frac{1}{2}$
- D)  $\frac{5}{12}$

**Quiz**

You roll a **fair six-sided die twice**.

Let:

- $X_1$ : result of the **first** roll
- $X_2$ : result of the **second** roll

Let event  $A$  be defined as:

$$A = \{X_1 < X_2\}$$

What is the probability of event  $A$ ?

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- A)  $\frac{15}{36}$
- B)  $\frac{21}{36}$
- C)  $\frac{30}{36}$
- D)  $\frac{6}{36}$

**Solved Problem**

You face **3 doors**: one hides a car, the other two hide goats.

You pick a door.

The host, who knows where the car is, opens **one of the other two doors** and shows a **goat**.

Then the host offers: “**Do you want to switch to the remaining unopened door?**”

**Question:** Is it to your advantage to switch?

**Solution**

Fix the car behind **Door 1**

| Initial pick | Host opens (must show a goat) | If you switch, you go to       | Result |
|--------------|-------------------------------|--------------------------------|--------|
| Door 1       | Door 2 or 3                   | the other unopened door (goat) | Lose   |
| Door 2       | Door 3                        | Door 1 (car)                   | Win    |
| Door 3       | Door 2                        | Door 1 (car)                   | Win    |

$$P(\text{win by switching}) = 2/3.$$

## Engelsk – dansk oversættelse

| English      | Dansk                          |
|--------------|--------------------------------|
| Set          | Mængde                         |
| Element      | Element                        |
| Subset       | Delmængde                      |
| Union        | Forening                       |
| Intersection | Fællesmængde                   |
| Complement   | Komplement                     |
| Cardinality  | Kardinalitet / Antal elementer |
| Countable    | Tællelig                       |
| Uncountable  | Utællelig                      |
| Equal sets   | Lige mængder                   |

## Engelsk – dansk oversættelse

| English               | Dansk                  |
|-----------------------|------------------------|
| Probability           | Sandsynlighed          |
| Random experiment     | Stokastisk eksperiment |
| Sample space          | Udfaldsrum             |
| Event                 | Hændelse               |
| Outcome               | Udfald                 |
| Axioms of probability | Sandsynlighedsaksiomer |
| Relative frequency    | Relativ frekvens       |